PROBLEM 41

Satellite Solar Panel: Composition of Rotations

Original statement · Problems.pdf, p. 29

Relative Kinematics

The angle θ describing the rotation of the solar panels of a reconnaissance satellite about the body-fixed axis x is an arbitrary function of time. The satellite spins about the z axis at the constant angular rate ω. Derive expressions for the absolute velocity and acceleration of point B in the figure relative to an inertial reference frame centered in A.

(Taken from Ginsberg, "Engineering Dynamics", Cambridge University Press, 2008.)



Satellite body spinning about z with a solar panel rotated by the angle θ about the body-fixed x axis, points A and B, dimensions H and W

Figure: Satellite body spinning about z with a solar panel rotated by the angle θ about the body-fixed x axis, points A and B, dimensions H and W (Problems.pdf, p. 29).

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a satellite body spins about its z axis at constant rate ω; a solar panel turns about the body-fixed x axis through the prescribed angle $\theta(t)$. Point B is a corner of the panel.
- **Why relative motion.** The motion of B is the superposition of two simple motions: in the body frame S_1, B just describes a circle of radius W about the x axis; and S_1 itself just spins about z. Notes Eqs. 3.26–3.27 combine the two without writing the inertial position explicitly. A second route uses the panel frame S_2, in which B is fixed, and the composition of angular velocities (Notes Eqs. 3.13–3.15).
- **Hypotheses:**

1. The frame centred at A with non-rotating axes is inertial, so the centre A of the satellite is fixed: $\mathbf{v}_0^A = \mathbf{a}_0^A = \mathbf{0}$.
2. Body and panel are rigid; ω is constant; $\theta(t)$ is an arbitrary (known) function.
3. **Geometry (read from the figure):** B is at distance H from A measured along x, and at distance W from the x axis; θ is measured about $+x$ from the body axis y, so for $\theta = 0$ the segment from the hinge axis to B points along $+y$.

- **Degrees of freedom:** none to determine; the motion is prescribed by ωt and $\theta(t)$ (kinematics only).

Symbol	Meaning	SI unit
H	axial distance from A to B (along x)	m
W	distance from B to the hinge axis x	m
ω	constant spin rate of the satellite about z	rad/s
$\theta(t)$	panel angle about x	rad

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 = \{A; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Body frame:** $S_1 = \{A; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ (x, y, z of the figure), a simple rotation of angle ωt about $\mathbf{k}_0 = \mathbf{k}_1$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1, \qquad \boldsymbol{\alpha}_{10} = \mathbf{0}$$

- **Panel frame:** $S_2 = \{C; \mathbf{i}_2, \mathbf{j}_2, \mathbf{k}_2\}$, where $C = A + H\mathbf{i}_1$ is the point of the hinge axis at the abscissa of B; simple rotation of angle θ about $\mathbf{i}_1 = \mathbf{i}_2$:

$$[{}_1R_2] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \qquad \boldsymbol{\omega}_{21} = \dot\theta\,\mathbf{i}_1$$

- **Checks:** $\det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$ and $\det[{}_1R_2] = \cos^2\theta + \sin^2\theta = 1$; the columns of each matrix are orthonormal, so $RR^T = I$; hence $[{}_0R_2] = [{}_0R_1][{}_1R_2]$ is also a proper rotation (Notes Eq. 1.16).
- **Position of B.** $\mathbf{r}_{CB} = W\,\mathbf{j}_2$ is fixed in S_2; with the second column of $[{}_1R_2]$, $\mathbf{j}_2 = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$, so

$$\mathbf{r}_1^B = \mathbf{AC} + \mathbf{CB} = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1$$

PHASE 3

Step-by-Step Mathematical Deduction

1. Relative motion of B in the body frame S_1

Pedagogical Justification: the body frame rotates at constant rate with fixed origin, so Notes Eq. 3.27 keeps only three terms; the relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant.

With the chain rule $\frac{d}{dt}\cos\theta = -\dot\theta\sin\theta$, $\frac{d}{dt}\sin\theta = \dot\theta\cos\theta$, and the product rule $\frac{d}{dt}(\dot\theta\sin\theta) = \ddot\theta\sin\theta + \dot\theta^2\cos\theta$, $\frac{d}{dt}(\dot\theta\cos\theta) = \ddot\theta\cos\theta - \dot\theta^2\sin\theta$:

$$\mathbf{v}_1^B = \left.\frac{d\mathbf{r}_1^B}{dt}\right|_1 = -W\dot\theta\sin\theta\,\mathbf{j}_1 + W\dot\theta\cos\theta\,\mathbf{k}_1$$

$$\mathbf{a}_1^B = \left.\frac{d\mathbf{v}_1^B}{dt}\right|_1 = -W\left(\ddot\theta\sin\theta + \dot\theta^2\cos\theta\right)\mathbf{j}_1 + W\left(\ddot\theta\cos\theta - \dot\theta^2\sin\theta\right)\mathbf{k}_1$$

(circular motion of radius W about the x axis: tangential part $W\ddot\theta\,\mathbf{k}_2$ and normal part $-W\dot\theta^2\,\mathbf{j}_2$).

2. Absolute velocity of B

> **Pedagogical Justification:** Notes Eq. 3.26 (Slide 24), $\mathbf{v}_0^B = \mathbf{v}_0^A + \mathbf{v}_1^B + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^B$, with $\mathbf{v}_0^A = \mathbf{0}$.

Cross products: $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1, \mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B = \omega \mathbf{k}_1 \times (H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1) = H\omega\,\mathbf{j}_1 - W\omega\cos\theta\,\mathbf{i}_1$$

Adding $\mathbf{v}_1^B$ component by component:

$$\boxed{\mathbf{v}_0^B = -W\omega\cos\theta\,\mathbf{i}_1 + \left(H\omega - W\dot{\theta}\sin\theta\right)\mathbf{j}_1 + W\dot{\theta}\cos\theta\,\mathbf{k}_1}$$

which agrees with the official key (Problems.pdf, p. 76). The components are given in the body basis $\mathcal{B}_1$; in $\mathcal{B}_0$ they follow from $\{\mathbf{v}_0^B\}_0 = [{}_0R_1]\{\mathbf{v}_0^B\}_1$.

3. Absolute acceleration of B

> **Pedagogical Justification:** Notes Eq. 3.27 (Slide 25) with $\mathbf{a}_0^A = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$: $\mathbf{a}_0^B = \mathbf{a}_1^B + \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^B$.

Centripetal term (reusing $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B$ from step 2):

$$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = \omega \mathbf{k}_1 \times (-W\omega\cos\theta\,\mathbf{i}_1 + H\omega\,\mathbf{j}_1) = -W\omega^2\cos\theta\,\mathbf{j}_1 - H\omega^2\,\mathbf{i}_1$$

i.e. $-\omega^2 \mathbf{r}_\perp$ with $\mathbf{r}_\perp = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1$.

Coriolis term:

$$2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^B = 2\omega \mathbf{k}_1 \times \left(-W\dot{\theta}\sin\theta\,\mathbf{j}_1 + W\dot{\theta}\cos\theta\,\mathbf{k}_1\right) = 2W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$$

Sum, collecting the $\mathbf{i}_1, \mathbf{j}_1$ and $\mathbf{k}_1$ components of $\mathbf{a}_1^B$, the centripetal term and the Coriolis term:

$$\boxed{\mathbf{a}_0^B = \left(2W\omega\dot{\theta}\sin\theta - H\omega^2\right)\mathbf{i}_1 - \left(W\ddot{\theta}\sin\theta + W\dot{\theta}^2\cos\theta + W\omega^2\cos\theta\right)\mathbf{j}_1 + \left(W\ddot{\theta}\cos\theta - W\dot{\theta}^2\sin\theta\right)\mathbf{k}_1}$$

which agrees with the official key (Problems.pdf, p. 76).

4. Second route: panel frame and composition of angular velocities

> **Pedagogical Justification:** B is fixed in S_2, so $\mathbf{v}_2^B = \mathbf{a}_2^B = \mathbf{0}$ and Notes Eqs. 3.26–3.27 applied to S_2 (origin C) need only $\boldsymbol{\omega}_{20}$, $\boldsymbol{\alpha}_{20}$ and the motion of C. Angular velocities add (Notes Eqs. 3.13–3.15, Slide 19); angular accelerations do not simply add.

$$\boldsymbol{\omega}_{20} = \boldsymbol{\omega}_{21} + \boldsymbol{\omega}_{10} = \dot{\theta}\,\mathbf{i}_1 + \omega\,\mathbf{k}_1$$

Its derivative in S_0 by the Coriolis theorem (Notes Eq. 3.8), with $\frac{d}{dt}\big|_1\left(\dot{\theta}\,\mathbf{i}_1 + \omega\,\mathbf{k}_1\right) = \ddot{\theta}\,\mathbf{i}_1$ (ω constant):

$$\boldsymbol{\alpha}_{20} = \left.\frac{d\boldsymbol{\omega}_{20}}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times\boldsymbol{\omega}_{20} = \ddot{\theta}\,\mathbf{i}_1 + \omega\mathbf{k}_1\times\dot{\theta}\,\mathbf{i}_1 = \ddot{\theta}\,\mathbf{i}_1 + \omega\dot{\theta}\,\mathbf{j}_1$$

Motion of C (fixed in S_1 at $H\mathbf{i}_1$): $\mathbf{v}_0^C = \omega\mathbf{k}_1\times H\mathbf{i}_1 = H\omega\,\mathbf{j}_1$, $\mathbf{a}_0^C = \omega\mathbf{k}_1\times H\omega\,\mathbf{j}_1 = -H\omega^2\,\mathbf{i}_1$. With $\mathbf{r}_2^B = \mathbf{CB} = W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1$:

$$\boldsymbol{\omega}_{20}\times\mathbf{r}_2^B = \dot{\theta}W\cos\theta\,\mathbf{k}_1 - \dot{\theta}W\sin\theta\,\mathbf{j}_1 - \omega W\cos\theta\,\mathbf{i}_1$$

so $\mathbf{v}_0^B = \mathbf{v}_0^C + \boldsymbol{\omega}_{20}\times\mathbf{r}_2^B$ reproduces the boxed velocity. For the acceleration, $\mathbf{a}_0^B = \mathbf{a}_0^C + \boldsymbol{\alpha}_{20}\times\mathbf{r}_2^B + \boldsymbol{\omega}_{20}\times(\boldsymbol{\omega}_{20}\times\mathbf{r}_2^B)$:

$$\boldsymbol{\alpha}_{20}\times\mathbf{r}_2^B = \left(\ddot{\theta}\,\mathbf{i}_1 + \omega\dot{\theta}\,\mathbf{j}_1\right)\times\left(W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1\right) = W\ddot{\theta}\cos\theta\,\mathbf{k}_1 - W\ddot{\theta}\sin\theta\,\mathbf{j}_1 + W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$$

$$\boldsymbol{\omega}_{20} \times \left(\boldsymbol{\omega}_{20} \times \mathbf{r}_2^B\right) = \left(\dot\theta\,\mathbf{i}_1 + \omega\,\mathbf{k}_1\right) \times \left(-\omega W\cos\theta\,\mathbf{i}_1 - \dot\theta W\sin\theta\,\mathbf{j}_1 + \dot\theta W\cos\theta\,\mathbf{k}_1\right)$$

$$= -\dot\theta^2 W\sin\theta\,\mathbf{k}_1 - \dot\theta^2 W\cos\theta\,\mathbf{j}_1 - \omega^2 W\cos\theta\,\mathbf{j}_1 + \omega\dot\theta W\sin\theta\,\mathbf{i}_1$$

(using $\mathbf{i}_1 \times \mathbf{j}_1 = \mathbf{k}_1, \mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1, \mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$). Adding $\mathbf{a}_0^C = -H\omega^2\,\mathbf{i}_1$: the $\mathbf{i}_1$ component is $2W\omega\dot\theta\sin\theta - H\omega^2$, the $\mathbf{j}_1$ component $-W(\ddot\theta\sin\theta + \dot\theta^2\cos\theta + \omega^2\cos\theta)$, the $\mathbf{k}_1$ component $W(\ddot\theta\cos\theta - \dot\theta^2\sin\theta)$: the same boxed result. Note that the Coriolis term of route 1, $2W\omega\dot\theta\sin\theta\,\mathbf{i}_1$, appears in route 2 split in two equal halves, one from $\boldsymbol{\alpha}_{20}$ (the term $\omega\dot\theta\,\mathbf{j}_1$ that makes angular accelerations not add) and one from the double cross product.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Reading of the terms.** $-H\omega^2\mathbf{i}_1 - W\omega^2\cos\theta\,\mathbf{j}_1$ is the centripetal acceleration of the spin; $W\ddot\theta$ and $W\dot\theta^2$ terms are the tangential and normal accelerations of the panel rotation; $2W\omega\dot\theta\sin\theta\,\mathbf{i}_1$ is the Coriolis coupling of the two rotations, present only when the panel moves ($\dot\theta \neq 0$) with a velocity component normal to the spin axis.
- **Limits:** $\dot\theta = 0$ (panel locked) gives the rigid-body motion $\mathbf{v}_0^B = \boldsymbol{\omega}_{10} \times \mathbf{r}_1^B, \mathbf{a}_0^B = -\omega^2\mathbf{r}_\perp$; $\omega = 0$ (no spin) leaves the circular motion of radius W about the hinge.
- **Dimensions:** every velocity component is of the form $[W\omega] = [H\omega] = [W\dot\theta] = \mathrm{m\,s^{-1}}$; every acceleration component $[W\omega\dot\theta] = [H\omega^2] = [W\ddot\theta] = [W\dot\theta^2] = \mathrm{m\,s^{-2}}$.
- **Verification (sympy, `verify_P41.py`):** $[{}_0R_1]$, $[{}_1R_2]$, $[{}_0R_2]$ are proper rotations; the inertial first and second derivatives of $[{}_0R_1]\{\mathbf{r}_1^B\}$ equal the Notes Eq. 3.26 and 3.27 expressions; both routes give exactly the official key.
- **Cross-check with Topic 1.** The Topic 1 solution (Poisson formula on the body basis) gives the same velocity and acceleration; no discrepancy with the official key.

Source: Problems.pdf (PDF page 29); key on PDF page 76